EC015
Chemistry
Semester 1
Session 2026/2027
2 hours

EC015
Kimia
Semester 1
Sesi 2026/2027
2 jam

FINAL TOUCH

CHEMISTRY

2 hours

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INSTRUCTIONS TO CANDIDATE:

This question paper consists of **6 questions**. Answer all questions.

Maximum marks awarded are shown in brackets at the end of each question or section.

The use of electronic calculator is permitted.

RELATIVE ATOMIC MASSES OF SELECTED ELEMENTS

Element	Symbol	Proton Number	Relative Atomic Mass
Aluminium	Al	13	27.0
Silver	Ag	47	107.9
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Gold	Au	79	197.0
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Iron	Fe	26	55.9
Fluorine	F	9	19.0
Phosphorus	P	15	31.0
Helium	He	2	4.0
Mercury	Hg	80	200.6
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Cadmium	Cd	48	112.4
Potassium	K	19	39.1
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Chlorine	Cl	17	35.5
Cobalt	Co	27	58.9
Krypton	Kr	36	83.8
Chromium	Cr	24	52.0
Copper	Cu	29	63.6
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganese	Mn	25	54.9
Sodium	Na	11	23.0
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Platinum	Pt	78	195.1
Lead	Pb	82	207.2
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Selenium	Se	34	79.0
Cerium	Ce	58	140.1
Cesium	Cs	55	132.9
Silicon	Si	14	28.1
Scandium	Sc	21	45.0
Tin	Sn	50	118.7
Antimony	Sb	51	122.0
Strontium	Sr	38	87.6
Sulphur	S	16	32.1
Uranium	U	92	238.0
Tungsten	W	74	183.8
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES

Ionisation constant for water at 25 ºC	K_w	=	1.00×10^{-4} mol^2 dm^{-6}
Molar volume of gases	V_m	=	22.4 dm^3 mol^{-1} at STP
		=	24 dm^3 mol^{-1} at room temperature
Speed of light in a vacuum	c	=	3.0×10^8 m s^{-1}
Specific heat of water		=	4.18 kJ kg^{-1} K^{-1}
		=	4.18 J g^{-1} K^{-1}
		=	4.18 J g^{-1} ºC^{-1}
Avogadro's number	N_A	=	6.02×10^{23} mol^{-1}
Faraday constant	F	=	9.65×10^4 C mol^{-1}
Planck's constant	h	=	6.6256×10^{-34} J s
Rydberg constant	R_H	=	1.097×10^7 m^{-1}
		=	2.18×10^{-18} J
Ideal gas constant	R	=	8.314 J mol^{-1} K^{-1}
		=	0.08206 L atm mol^{-1} K^{-1}
Density of water at 25ºC	ρ	=	1 g cm^{-3}
Freezing point of water		=	0.00 ºC
Vapour pressure of water at 25ºC	P_{water}	=	23.8 torr

UNIT AND CONVERSION FACTOR

Volume	1 liter = 1 dm^3 1 mL = 1 cm^3
Energy	1 J = 1 kg m^2 s^{-2} = 1 N m = 10^7 erg 1 calorie = 4.184 Joule 1 eV = 1.602 x 10^{-19} J
Pressure	1 atm = 760 mmHg = 760 torr = 101.325 kPa = 101325 N m^{-2}
Others	1 Faraday (F) = 96500 coulomb 1 newton (N) = 1 kg m s^{-2}

1. (a) The mass spectrum of Neon is shown in FIGURE 1. The average atomic mass of Neon is 20.18 amu. Determine x and y in $^{21}_{10}Ne$ and $^{22}_{10}Ne$.

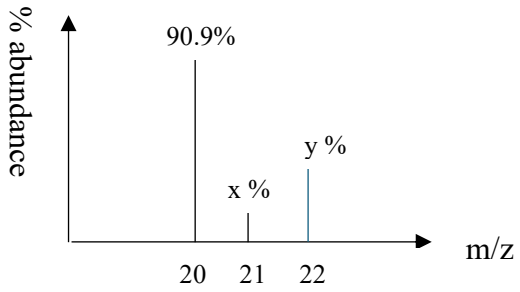


FIGURE 1 [3M]

(b) A 50 mL of saturated NaOH containing 52% by mass with density of 1.48 gmL^{-1} is used to prepare a 0.1 M NaOH solution.

(i) Determine the initial concentration of the saturated NaOH solution.
(ii) Determine the amount of water required to prepare the 0.1 M NaOH solution.

[8M]

(c) A 25.0 cm^3 sample of an acidified solution containing ethanedioic acid, $H_2C_2O_4$, was titrated against a standard solution of potassium dichromate (VI), $K_2Cr_2O_7$ with a concentration of 0.02 mol dm^{-3}. Dichromate (VI) ion, $Cr_2O_7^{2-}$ undergoes reduction as it is converted into chromium (III) ions Cr^{3+}, whereas the oxalate ion $C_2O_4^{2-}$ from the acid is oxidised to carbon dioxide gas CO_2. The titration required exactly 22.50 cm^3 of the $K_2Cr_2O_7$ solution to reach a sharp end point.

(i) Balance the redox equation using ion-electron method.
(ii) Determine the molar concentration, in mol dm^{-3} of the ethanedioic acid solution.

[5M]

(d) When excess ethanoic acid, CH_3COOH reacts with calcium hydroxide, $Ca(OH)_2$, salt and water is produced. Determine the amount of reactant left over (in mol) if 20 mL of 4 M ethanoic acid reacted with 0.76 g of calcium hydroxide.

[5M]

2. (a) The line spectrum of hydrogen atom in the ultraviolet region is shown in **FIGURE 2**.

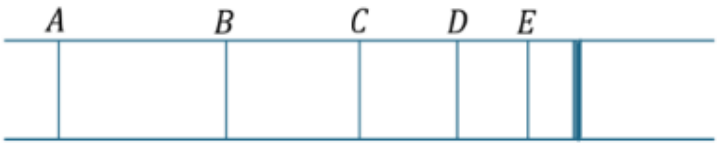


FIGURE 2

(i) State the electron transition that will produce the line C.
(ii) Calculate the energy of the photon that produces line B?
(iii) Which of the labelled lines has the longest wavelength?

[5M]

(b) An ion Y^{3+} has 28 electrons and 38 neutrons

(i) Write the electronic configuration of atom Y.
(ii) Draw the shapes of orbitals occupied by the valence electrons of Y.
(iii) Write a set of quantum numbers for the fifth electrons in Y^{3+} ion.
(iv) Determine how many electron(s) exist in ion Y^{3+} when angular momentum quantum number, l= 0.

[5M]

3. (a) ClF_3 (Chlorine Trifluoride) and ClF_4^- (tetrafluoridochlorate (III) anion) are interhalogen species where the central chlorine atom exhibits hypervalency, making them notable exceptions to the octet rule.

(i) Draw the Lewis structure for both molecules.
(ii) Name the molecular geometry for both molecules.
(iii) Explain why do species like ClF_3 and ClF_4^- act as exceptions to the octet rule?
(iv) Illustrate the hybridisation process of chlorine in ClF_4^-.

[8M]

(b) Thiocyanate ion, SCN^- has numerous applications in serving as the starting materials for the synthesis of certain dyes and in pharmaceuticals.

(i) Draw all the resonance structures for thiocyanate ion, SCN^-.
(ii) Based on the formal charge concept, determine the most plausible structure, and give your reason.

[6M]

(c) H_2S and H_2O have the same molecular shape but boiling point of H_2O is higher than H_2S. Explain.

[3M]

4 (a) Phosphorus vapour exists as a tetraatomic molecule, P_4 and behaves as an ideal gas under certain conditions. Given the density of phosphorus vapour at 310°C and 775 mmHg is 2.64 g/L.

(i) What is the molecular mass of phosphorus under these conditions?
(ii) The phosphorus initially occupies a volume of 2.5 L. If the temperature is increased to 400^0C while keeping the pressure constant, what will be the new volume of the phosphorus vapour?
(iii) At which condition P_4 behaves as a real gas? [6M]

(b) Unknown X is a compound where its solid state is denser than its liquid state. The triple point and critical point of unknown X is given in the following Table 1.

	Pressure (atm)	Temperature (^{0}C)
Triple point	5.1	-57
Critical point	73	31

Table 1

(i) Sketch a label phase diagram of unknown X by using the information given.
(ii) Explain why there is a difference in the slope of the equilibrium line of solid-liquid in the phase diagram of water.

[3M]

5 The equation of a stimulated photosynthesis reaction is represented by:

$6CO_2\ (g) + 6H_2O\ (g) \rightleftharpoons C_6H_{12}O_6(s) + 6O_2(g) \qquad \Delta H = +ve$

At 31°C, the following equilibrium concentrations were found:

$[H_2O] = 7.91 \text{ X } 10^{-2}$ M, $[CO_2] = 9.30 \text{ X } 10^{-1}$ M, $[O_2] = 2.40 \text{ X } 10^{-2}$ M

(a) Calculate the equilibrium constant K_p for this reaction. [3M]
(b) Determine the initial concentration of CO_2 and H_2O assuming there is no presence of product at the starting of the reaction. [5M]
(c) What will happen to K_p if temperature is increased? [1M]

6 (a) The table shows the acid-dissocation constant, K_a for several weak acids.

Acid	K_a
HCN	$4.9 \text{ x } 10^{-10}$
C_6H_5COOH	$6.5 \text{ x } 10^{-5}$

i. Give the formula of the conjugate base for each acid.
ii. Which one is the stronger acid? Why? [4M]

(b) In an experiment, a student was required to titrate 200 mL of 0.30 M ammonia,NH_3 against 100 mL of 0.2 M hydrochloric acid. ($K_b\ NH_3 = 1.8 \text{ x } 10^{-5}$)

i. Write the chemical equations for the reaction above.
ii. Determine the pH and nature of the final solution.

[7 M]

(c) The solubility product, K_{sp} of Ag_2SO_4 is $1.50 \text{ x } 10^{-5}$. Calculate the solubility of the salt in 1.0 M H_2SO_4 solution.

[3M]